

Nicholas Belev

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EDUCATION

McGill University, Montreal, QC

Graduated: May 2026

B.Sc. Computer Science & Artificial Intelligence; Minors in Mathematics, Finance

GPA: **3.91 / 4.00**

Key Coursework: Machine Learning, Algorithm Design, Quant Risk Management, Statistics, Portfolio Theory.

EXPERIENCE

Santander Bank, *Data Analytics Intern*

Boston, MA

Jun 2025 – Aug 2025

- Led the firm's first AI (LLM) audit by designing adversarial prompt test suites and statistical performance metrics; automated API testing to evaluate and remediate model risk, prompt injection defenses, and PII handling.
- Built a Power BI dashboard backed by Oracle SQL and DAX to forecast audit workload and support resource allocation across a **150+ person** department.
- Refactored Anti-Money Laundering (AML) tool by optimizing TF-IDF vectorization and fuzzy matching pipelines, achieving a **3x runtime improvement**.

Aspequity, *Chief Technology Officer & Co-Founder*

Boston, MA

Sep 2025 – Present

- Co-founded a private markets startup building forecasting software for private equity cash flows and valuations, serving paying institutional clients.
- Developed C++ forecasting platform; leading deployment of secure cloud API for client portfolio analytics access.
- Built data pipelines for private equity LP data, storing commitments, distributions, NAV, TVPI, and IRR metrics in PostgreSQL.

Northfield, *Quantitative Analyst & Software Engineer*

Boston, MA

May 2024 – Sep 2024

- Developed portfolio construction and optimization analytics in Python supporting long/short equity strategies for a **\$65B AUM** asset manager.
- Built SQL and Python pipelines for factor attribution, volatility monitoring, and portfolio performance analysis.
- Migrated **16,000+** legacy factor model files into PostgreSQL and deployed a FastAPI service enabling **sub-10 ms** risk queries for portfolio managers.

PROJECTS

Portfolio Project (Custom ETF), *Team Lead*

Sep 2025 – Dec 2025

- Led a 6-person team in a semester-long project to design a custom equity portfolio; project received the **highest overall evaluation**.
- Built a Python analytics pipeline using **10+ years** of daily return data and the **Fama–French multifactor model** to estimate time-varying betas and alpha via **60-month rolling OLS regressions**.
- Performed systematic vs. idiosyncratic risk decomposition and reduced idiosyncratic risk to **< 20%** through factor-aware portfolio construction and exposure constraints.

Financial Risk Modeling & VaR Forecasting (on [GitHub](#))

Mar 2026 – Apr 2026

- Modeled daily returns of a financial asset (Uranium ETF) using **ARMA-GARCH(1,1)** with Student-t innovations to capture volatility clustering and kurtosis.
- Forecasted Value-at-Risk (VaR) and Expected Shortfall (ES) at the **95%** and **99%** levels using historical simulation, EVT (POT/GPD), and parametric methods.
- Built rolling backtesting pipeline (**1000-day** window) to determine violations and evaluate tail risk forecasts.

Othello AI Agent Competition (on [GitHub](#))

Oct 2024 – Dec 2024

- Implemented an Othello-playing AI using alpha–beta pruning, iterative deepening minimax, and Zobrist hashing under **2s** time and **500MB** memory constraints.
- Designed an evaluation function incorporating stability, mobility, and positional control.
- Ranked in the **top 10 out of 200 teams** in tournament; achieved **100%** win rate against greedy agents, **99%** against stochastic agents, and **80%** win frequency against top peer implementations.

TECHNICAL SKILLS

Programming: Python, Java, C/C++, OCaml, Solidity (Ethereum)

Data & Analytics: SQL (PostgreSQL, Oracle, MS SQL Server), R, Pandas, NumPy, scikit-learn

Backend & Systems: FastAPI, REST APIs, Docker, Linux/Unix, Bash

Tools: Excel, Power BI, AWS (EC2), Git, LaTeX